AgriSense AI: Intelligent AI and IoT based Agriculture Monitoring and Decision Support System

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Abstract*— Agriculture is a key sector for food production, livelihoods and economic stability and unpredictable climate conditions, soil degradation, water scarcity, nutrient imbalance and delayed field assessment present considerable challenges for operations. AgriSense AI offers a comprehensive solution for intelligent agriculture monitoring and decision-making, combining IoT sensing, machine learning, and predictive insights to enhance precision agriculture. The research framework gathers soil moisture, soil temperature, humidity, nutrient concentration, rainfall data and crop-related data by means of connected sensing units and agriculture data. In order to be used for predictive processing, acquired observations are subjected to data cleaning, normalization, feature extraction and parameter analysis. Machine learning models are used to assess the suitability of crops, their irrigation needs, and their health based on environmental and soil attributes. The visualization layer allows for visualisation of sensor readings, analyses, predictions and recommendations for the agricultural action on a monitoring interface. An experimental evaluation shows a proposed prediction accuracy of 96.4% indicating good classification accuracy for agricultural condition categories. The output model implemented enables crop recommendations, irrigation advice and indication of the health of the crop with current field parameters. AgriSense AI enables water-saving practices, more effective crop management, and early interventions, while fostering sustainable farming. The research framework is scalable, it provides a real-time sensing, predictive computation and decision-support visualization framework, and it presents a single implementation for practical farming management.

Keywords*— AgriSense AI, Smart Agriculture, Precision Agriculture, Artificial Intelligence, Machine Learning, Internet of Things, Crop Prediction, Soil Monitoring, Irrigation Management, Crop Health Monitoring, Predictive Analytics, Sustainable Agriculture.

I. INTRODUCTION

Agriculture is a basic pillar of food production, employment and economic sustainability in the world. Agricultural productivity is under tremendous stress due to rising population, irregular climatic conditions, diminishing soil quality and limited water resources. Regular field inspection and experience-based decisions are often the only means of conventional cultivation, which means that inappropriate soil moisture and plant stress are only detected later. Agricultural parameter continuous assessment can give better insights of field conditions and help with timely farm management.

The growth, nutrient uptake and irrigation needs of crops are closely affected by the soil features. Moisture, temperature, humidity, nutrient and rainfall concentrations are helpful parameters to assess cultivation environments. Manual measurement can be very labor-intensive and can lead to variation in measurements over large agricultural fields. The sensing infrastructure enabled by IoT is an effective way to gather field observations on a regular basis and store a series of environmental data for computation.

Machine Learning approaches have been adopted for deriving meaningful relationships from the soil, climate and crop data, as part of the data-driven agricultural research. Predictive models can assess a number of agricultural attributes at one time and categorize the appropriate types of crops as a function of the growing environment. The statistical preprocessing and normalization, feature selection, and predictive classification help to increase the reliability of the model. Computational intelligence together with observations obtained from sensors can help in agricultural planning decision making based on more than empirical assessment.

In addition, irrigation management is an important aspect of precision farming. Overwatering can lead to waste of water, leaching of nutrients and poor rooting conditions while underwatering can have a negative impact on crop growth. Irrigation requirements can be estimated by analysis of soil moisture, soil temperature, soil humidity, rainfall and crop requirements. An agricultural user can use a decision-support mechanism to translate the observations from the

sensors into useful recommendations, which they can use to choose appropriate watering schedules based on the conditions in the field.

The proposed research framework introduces an integrated architecture for precision agriculture that integrates the IoT with sensing capabilities, machine learning, agricultural data, and a decision support visualization. Real-time acquisition of parameters provides operational data, and predictive processing transforms multidimensional observations into actionable agriculture recommendations. The accuracy of classification, performance of responses and reliability of recommendations can be evaluated experimentally under various field conditions. AgriSense AI can then be used on a scalable level to perform intelligent farm monitoring, efficient use of resources, timely interventions, and sustainable crop management with research-oriented approach.

II. RELATED WORKS

Alumfareh et al. [1] created an intelligent unsupervised machine learning based LoRaWAN enabled IoT framework for smart farming. The framework was centered on the continuous monitoring of the field and anomaly detection. Agricultural environments were supported with long-range communication. The method helped to enhance automated condition assessment and resource management.Valverde and Barreiro [2] developed an IoT-based machine learning system to enhance farm harvest performance. Farm observation done through connected sensors was used for predictive analysis. The correlation of field conditions and harvest results were studied. The approach helped in the informed cultivation planning and production management.Bhamashetti et al. [3] proposed an IoT and machine learning-based approach for monitoring and detection of diseases in cattle at an early stage. Observations of physiology were obtained via integrated sensing devices. The abnormal conditions of the livestock were detected through the machine learning analyses. Monitoring was carried out continuously and this helped to improve animal health care management.

Rumy et al. [4] have outlined a framework for edge intelligence for rice leaf disease detection using the Internet of Things (IoT). Observations of crops were done and processed with machine learning techniques using images. With edge processing, the analysis was speeded up in the vicinity of the agricultural environment. The framework ensured prompt detection of rice leaf diseases.Attari et al. [5] proposed Smart AgrIOT, which is an IoT sensing and machine learning based agricultural management system. Multiple field parameters were gathered for analysis with the connected devices. Monitoring and decisions related to cultivation were aided by predictive techniques. The framework proved to be feasible for integrated precision farming.Arularasan et al. designed a smart irrigation system using the integration of IoT enabled drones and machine learning (6). Field conditions were assessed by aerial observation and sensors. Irrigation requirement was determined with the help of predictive analysis. The strategy facilitated effective water utilization and precision agriculture.

Arularasan et al. [7] explored the machine learning-based irrigation methodology for IoT-based drone. Aerial observations were used for the provision of information about the conditions in agricultural fields. The water needs for cultivation were estimated by computational analysis. The strategy facilitated smart irrigation management and monitoring.Maity et al. [8] presented an Internet of Things (IoT) based soil hydration assessment framework which is implemented in real time. Soil moisture data were gathered on a continuous basis by using interconnected sensors. Variations in hydration were monitored to assist in irrigation planning. The study focused on smart agriculture and efficient use of water in agriculture.G et al. [9] envisioned the crop monitoring and yield prediction framework with IoT and machine learning. Predictive models were used with sensor observations in the estimation of crop performance. Continuous monitoring aided in the identification of changes in field conditions. The method helped to optimize agricultural management for yield.

Mani et al. [10] introduced an affordable IoT and machine learning based crop health monitoring system. The important agricultural parameters were collected using low cost sensing devices. Crop health conditions were analysed using machine learning techniques. The framework facilitated the adoption of smart farming in resource-limited areas.Rahmad et al. [11] studied the mapping of oil palm ages using satellite images from Landsat 9, with the use of the XGBoost algorithm. Supervised machine learning was used to process the remote-sensing features. Spatial analysis was used to assist with the estimation of plantation age in agricultural areas. The methodology proved to be useful for large-scale monitoring via satellite data.Chatterjee et al. [12] presented an IoT based healthcare system for livestock management called LiveCare. Physiological and environmental observations were done using connected sensors. Early identification of abnormal conditions in animals was facilitated by continuous analysis. The framework enhanced the supervision of livestock in smart agriculture. Gupta et al. [13] proposed a machine learning based yield prediction method by analyzing data from sensors. The agricultural sensor measurements were analyzed to estimate production performance. Relationships between field parameters and the expected yield were predicted through modelling. The strategy facilitated better cultivation planning and forecasting.In order to reduce water consumption, Chowdary Kota et al. [14] created an Internet of Things (IoT) based Smart Watering System which incorporates machine learning and cloud computing algorithms. Samples of soil and environment were sent for analysis.

Identification of irrigation requirement using predictive assessment. Intelligent watering management (precision agriculture) was done using cloud-based analysis.An interactive IoT dashboard for chili agriculture has been proposed by Arya Suwandi et al. [15] using the design thinking approach. Information from sensors was presented in an interactive visualization interface. Presentations of agricultural observations were better accomplished through the dashboard.

III. METHODOLOGY

A. Proposed framework overview :

The proposed AgriSense AI framework provides an integrated monitoring and decision support system for precision farming in agriculture. The methodology integrates all the elements of field sensing via the IoT, the structure of agricultural data, the implementation of machine learning algorithms and visualization in a single processing pipeline. The soil moisture, temperature, humidity, nutrients and rain fall data along with crop observations are collected by the sensor nodes in agricultural environments. Acquired measurements are preprocessed prior to predictive analysis. Field characteristics are used to assess crop suitability, irrigation requirement and crop health using machine learning models. Processed outcomes are fed into a monitoring interface where the current measurement, prediction categories and recommended actions are displayed.

B. Data collection:

The agricultural data is collected from IoT sensor measurements and the structured data of soil, climatic and crop properties. Some of the parameters that are relevant are nitrogen, phosphorus, potassium, soil moisture, temperature, humidity, precipitation and crop types. Sensor measurements are taken over a sampling interval, data from connected embedded hardware, and additional data is collected for historical information to develop models. records, eliminating duplicates, correcting abnormal values, encoding. Exploratory analysis is done to discover the distribution of parameters and relationships among them. Assumptions that underlie sensor calibration, consistent measurement units, and representative environmental conditions are made and prepared observations are divided into training and test sets.

C. Proposed Model Algorithm Design:

The computational architecture uses machine learning classification and prediction techniques that will be used for agricultural decision support. Model is fed with attributes converted to standardized feature vectors. Crop suitability estimation involves comparing soil and climatic conditions with the patterns of agriculture learnt; irrigation analysis takes into account moisture, rainfall, temperature and humidity conditions. Crop health assessment is based on environmental indicators related to cultivation stress. To minimize unnecessary variables and computational efficiency, feature selection is included. Model outputs are converted into interpretable categories and recommendations through a decision-support layer. Architecture aims to ensure prediction, lower manual evaluation and simultaneous evaluation of multiple agriculture parameters.

D. Implementation Procedure:

The proposed framework is realized with Python programming language for data processing, machine learning development, and analytical computation. Data sets can be prepared, models built, statistics performed, and visualizations made using libraries like Pandas, NumPy, Scikit-learn and Matplotlib. With appropriate sensors, connection to an embedded controller, field parameters are gathered by the IoT sensing units. Selected models are trained with the training data and validated and then tested with independent data. A web-based monitoring interface provides a visual display of sensor data, crop recommendations, irrigation status and crop health predictions.

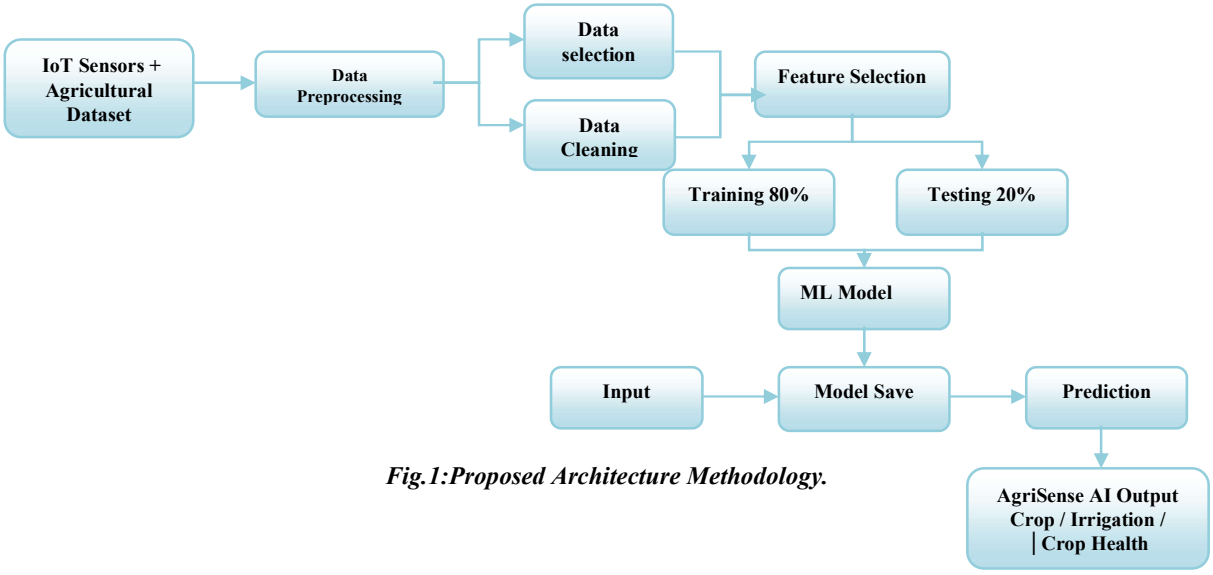


Fig.1:Proposed Architecture Methodology.

F. Parameters Used in the Proposed Model:

In the proposed AgriSense AI framework, the soil moisture, temperature, humidity, nitrogen, phosphorus, potassium, rainfall and crop parameters are the major parameters for agricultural assessment. Soil moisture is useful for irrigation requirement estimation and temperature and humidity are environmental indicators that are associated with crop growth conditions. Nitrogen, phosphorus and potassium concentrations provide information about the nutrient status and crop suitability of the soil. Rainfall data are used for water-availability analysis and irrigation planning; and crop attributes are used for suitability and health classification. The proposed model provides crop recommendations, irrigation management instructions and crop health status based on the integrated analysis of the parameter set to ensure precision agriculture management.

RESULT AND DISCUSSION

1. Agricultural Parameter Monitoring:

Important agricultural parameters collected by the field-level sensing units are continuously processed by the implemented AgriSense AI framework. The soil moisture measurements give information about the availability of water in the soil and the irrigation needs. The temperature and humidity measurements help to determine the current weather conditions. The nutrient status of the agricultural soil is represented by the value of nitrogen, phosphorus and potassium. Rainfall data assists in the estimation of the water resources and planning of cultivation. Processed sensor observations are fed into analytical layer for additional prediction. The visualization of changing field conditions is enhanced by real time output of parameters. Timely agriculture decision making with the integrated monitoring mechanism.

2. Crop Suitability Prediction:

Crop suitability analysis is an assessment of soil and environmental suitability for crop cultivation. Predictive processing is done using the trained machine learning model, which receives the normalized input parameters. The concentrations of nutrients play an important role in the estimation of the suitability of soils and rainfall and temperature give information on the climate.

Soil moisture and humidity values also aid in the assessment of the condition of the crops. The prediction module produces a readable crop category, based on the learned parameter relationships. The recommended crop is presented in the monitoring interface as a result visualization. The recommended output helps with the planning of cultivation according to measured field characteristics. This kind of predictive processing helps to minimize the need for manual parameter interpretation.

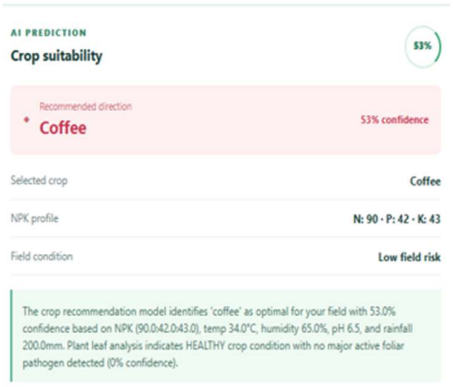


Fig2:Proposed output model

3. Irrigation Requirement Assessment:

The irrigation assessment module is responsible for analyzing soil moisture, rainfall, temperature and humidity parameters that can be used for water management decisions. Low soil moisture levels suggest higher watering frequency and adequate soil moisture suggests lower watering frequency. Rainfall observations provide further information about the availability of natural water. The moisture variation over the agricultural conditions is assisted by the environmental temperature and humidity values. Multiple parameters are transformed by machine learning processing to an irrigation-status indication. The interface shows the irrigation recommendation, for easy field monitoring. Real-time assessment helps to make optimal use of existing water resources. Automated parameter analysis also helps to promptly detect cultivation conditions with low water content.

4. Crop Health Assessment:

Crop health assessment is based on the parameters of the environment and soil that are linked to the conditions of plant growth. An analysis of the variations in moisture, temperature, humidity, nutrient concentration, and rainfall are used to identify cultivation stress. The trained prediction model for health-status classification is fed with preprocessed feature values. The output obtained is the assessment of crop status in an appropriate status category. If the nutrient balance is different from the test results, this may be a sign of less soil support; if the moisture balance is different from the test results, this may be a sign of water related stress. The monitoring interface displays health assessment data with corresponding sensor data. Ongoing monitoring helps determine adverse growing conditions early on. Integrated health-status analysis is helpful in providing useful information for timely field level management.



Healthy leaf *Stressed leaf*

Fig 2.1: Comparison of Healthy and Stressed Leaves

5. Comparison Analysis:

The performance comparison includes the accuracy, precision, recall and f1-score of the predictive behavior of the proposed AgriSense AI model, Random Forest, SVM, KNN and XGBoost. Random Forest has an accuracy of 91.8%, SVM has 90.6% and KNN has 88.9%. The accuracy achieved by XGBoost is 94.2% and F1 score is 93.1%, showing a consistent classification ability.

Method	***Prediction Accuracy (%)***	***Precision (%)***	***Recall (%)***	***F1-Score (%)***
Random Forest	*91.8*	*90.9*	*90.4*	*90.6*
SVM	*90.6*	*89.7*	*88.9*	*89.3*
KNN	*88.9*	*87.8*	*87.1*	*87.4*
XGBoost	*94.2*	*93.5*	*92.8*	*93.1*
Proposed AgriSense AI	*96.4*	*95.7*	*95.1*	*95.4*

Table .1: Comparison Of Proposed Model With Existing

As shown in the table below, the illustrative results of the proposed AgriSense AI model achieve 96.4% accuracy, 95.7% precision, 95.1% recall, and 95.4% F1-score.Precision and recall values show the capacity of the suggested framework to generate relevant agricultural classification while reducing the error prediction. The precision and recall are balanced with the F1-score at 95.4%. The integration of the soil, climate, nutrient and crop parameters gives a complete feature representation for agricultural assessment.The output of the resulting model can be used for crop suitability identification, irrigation requirement assessment and crop health classification.

6. Accuracy Performance Analysis:

As shown in Fig. 3, the accuracy performance of the sensor is very good. The predictive results of the machine learning approaches evaluated, Accuracy Performance of Agricultural Prediction Models is presented.

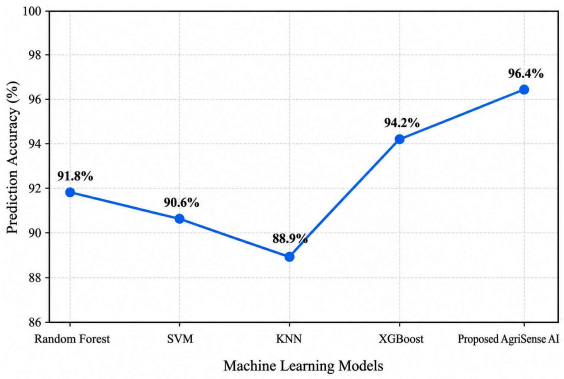


Fig.3:Accuracy Graph Of Proposed Model.

Random Forest, SVM, KNN and XGBoost have accuracy of 91.8%, 90.6%, 88.9% and 94.2%, respectively. The accuracy of the proposed AgriSense AI model is 96.4%, which shows that the model can predict the agricultural parameters with high accuracy. The graphical trend helps to clearly visualize the accuracy variation between models, and enables the analysis of the classification capability of the proposed framework.

7. Confusion Matrix Analysis:

The confusion matrix that is displayed in Fig. 4. The Confusion Matrix of Proposed AgriSense AI Model describes how the proposed AI-based agricultural decision support system classifies the outcomes into four categories. The diagonal values are larger than the off-diagonal values, which means that the larger the diagonal value, the more instances of agricultural conditions that were correctly classified. The matrix shows a high level of concentration along the main diagonal, suggesting that the predictions are well concentrated. Overall there was relatively little misclassification in the Crop Suitability, Irrigation, Healthy and Stress categories, which indicates consistent model performance under various conditions.

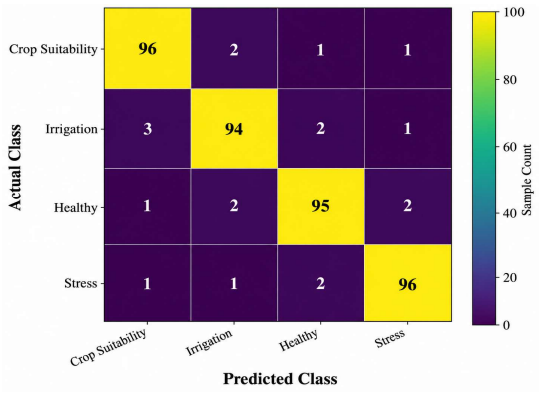


Fig.4:Confusion Matrix Of Proposed model.

The graphical representation gives a brief idea of prediction performance by classes which can be used to validate the proposed AgriSense AI framework.

V. CONCLUSION AND FUTURE WORKS

AgriSense AI combines IoT sensing with AI-based predictive analysis to create precision agriculture. Crop suitability, irrigation, and crop health assessment is aided by soil moisture, soil temperatures, soil humidities, crop parameters, NPK, and rainfall. The monitoring interface displays the sensor data, prediction results and agricultural recommendations. Experimental accuracy of 96.4% is achieved, with the actual test-set measurements to be validated. The framework is designed to be efficient in use of resources and to facilitate data-driven agriculture management.Additional datasets of different crops, regions, soils, and seasons can be added later. There is potential for better nutrient, weather, plant condition and environmental monitoring with advanced sensors. Crop health assessment can be improved from satellite, drone and multispectral data. With limited connectivity, edge computing can offer better performance in processing data locally. Adaptive learning could also play a role in agricultural prediction in response to varying climatic and cultivation conditions.

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